

It is important to note here that the feasible range of model parameters is also dependent upon the current financial regime. For example, sensitivity and parameter values during the financial crisis of 2008–2009 could be much different than during a low volatility regime. Analysts need to continuously evaluate what constitutes a feasible range for the parameters.

The process we used to determine the sensitivity of model results to these parameters is as follows:

1. Start with parameter a_1 .
2. Sets its value to $a_1 = 100$.
3. Solve the non-linear least squares model with $a_1 = 100$ and the above constraints on the other parameters.
4. Record the resulting parameter values and non-linear R^2 estimate e.g., $(a_1 = 100, a_2 = \hat{a}_2, a_3 = \hat{a}_3, a_4 = \hat{a}_4, b_1 = \hat{b}_1, NonR^2)$
5. Increase the value of a_1 (e.g., set $a_1 = 150$) and re-run the non-linear least squares regression, record the values, repeat until $a_1 = 1000$.
6. Repeat these steps for all the parameters., i.e., hold one parameter value constant and solve for the other four. Record results.
7. Plot and analyze the results.

We performed the above analysis for all feasible values of the parameters. For each parameter we plotted the specified parameter value and the non-linear R^2 from the best fit non-linear regression. For example, for $a_1 = 100$, the best fit non-linear R^2 was $R^2 = 0.23$. For $a_1 = 150$, the best fit non-linear R^2 was $R^2 = 0.38$, etc.

The results of our sensitivity analysis are shown in Figure 5.6a–e. Figure 5.6a shows the results for parameter a_1 . The graph shows R^2 increasing from 0.28 (at $a_1 = 100$) to a maximum value of $R^2 = 0.41$ (at $a_1 = 700$), and then decreasing again to $R^2 = 0.38$. If we look at Figure 5.6a we find that the best fit R^2 value varies very little between the values $a_1 = 600$ to $a_1 = 800$. The best fit equation is pretty flat between these values. This type of result is not unique to the I-Star model; it is in fact pretty common across most non-linear equations and is the reason we have been stressing the need to perform a thorough sensitivity analysis on the data.

Figure 5.6b–e illustrates the sensitivity analysis for a_2 through b_1 . Parameter a_2 has its best fit value at about $a_2 = 0.55$ and appears to have a range between $a_2 = 0.45$ to $a_2 = 0.65$. Parameter a_3 reaches its best fit at $a_3 = 0.75$ with a range of about $a_3 = 0.65$ to $a_3 = 0.80$. Parameter a_4 reaches its best fit at $a_4 = 0.45$ with a range of about $a_4 = 0.4$ to $a_4 = 1$. Parameter b_1 reaches its best fit at $b_1 = 0.92$ with a range of $b_1 = 0.87$ to